

Finding a Percent of a Quantity

Answer Key

Grades 6–7

Quick Answers

1. 8	4. 16	7. 20	10. 45, 48.6
2. 23	5. 6	8. 7, 28	
3. 21	6. 7.2	9. 8	

Curriculum

Level: Grades 6–7

6.RP.A.3.c – problems 1, 2, 3, 4, 5, 7, 9

7.RP.A.3 – problems 6, 8, 10

Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

5. *If they got 12: moved the decimal point one place and found ten percent instead of five percent*
6. *If they got 4.8: used ten percent instead of fifteen percent*
8. *If they got 15: took twenty dollars off the price instead of twenty percent of the price*
10. *If they got 64.8: added the eight percent tax to the original price instead of to the sale price*
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1. Bar model: 10% of 80 books. The bar stands for the whole collection, which is 100%, and it is cut into ten equal parts, so each part is $\frac{100}{10} = 10\%$ of the books. Each part must therefore hold $\frac{80}{10}$ books. Dividing the whole by ten is exactly what taking 10% does, which is why 10% is the easiest benchmark to find. So 10% of 80 = 8 books.

2. Bar model: 50% of 46 marbles. The bar is cut into two equal parts, and two equal parts of 100% are 50% each, so one part is half the marbles. Half of 46 is $\frac{46}{2}$. Splitting 46 into 40 + 6 makes the halving easy: 20 + 3. So 50% of 46 = 23 marbles.

3. Ratio table: 30% of 70. The table starts from 100% → 70. Dividing both columns by ten gives 10% → 7. Multiplying both columns by three gives 30% → 21. In one step the same reasoning is a multiplication: $30\% = \frac{30}{100} = 0.3$, and $0.3 \times 70 = 21$. Both routes agree, so 30% of 70 = 21.

4. Fraction shortcut: 25% of 64 songs. $25\% = \frac{25}{100} = \frac{1}{4}$, so a quarter of the playlist is wanted and the quickest route is a division: $\frac{64}{4} = 16$. Check with the decimal route: $25\% = 0.25$ and $0.25 \times 64 = 16$, the same value. So 25% of 64 = 16 songs.

5. Find the mistake: 5% of 120. Moving the decimal point one place to the left divides by ten, and dividing by ten is taking $\frac{1}{10}$, which is 10% — not 5%. That is why Dana got 12: she found 10% of 120. Since 5% is half of 10%, halve her result: $\frac{12}{2} = 6$. Check with the decimal: $5\% = 0.05$, and $0.05 \times 120 = 6$. The correct value is 6, not 12.

6. Restaurant tip: 15% of 48 dollars. Split the benchmark: 10% of 48 is 4.8, and 5% is half of that, 2.4. Adding the two parts gives $4.8 + 2.4 = 7.2$. The one-step route agrees: $15\% = 0.15$ and $0.15 \times 48 = 7.2$. To the nearest cent the tip is 7.20 dollars.

7. Concert hall: 8% of 250 seats. Write the percent as a decimal: $8\% = \frac{8}{100} = 0.08$. Then $0.08 \times 250 = 20$. Benchmark check: 10% of 250 is 25, and 8% must be a little less than that, which 20 is. So the number of empty seats is 20 seats.

8. Jacket on sale: 35 dollars, 20% off. The discount is a percent *of the price*, so the equation is $d = \frac{20}{100} \times 35$. Solving: $0.2 \times 35 = 7$, so the discount is 7 dollars. The sale price is what is left of the price after the discount comes off: $35 - 7 = 28$. Equivalently, paying 80% of the price gives $0.8 \times 35 = 28$, the same value. The sale price is 28 dollars. (A student who wrote 15 took twenty *dollars* off rather than twenty percent.)

9. An awkward percent: 12.5% of 64 grams. $12.5\% = \frac{12.5}{100} = \frac{1}{8}$, so the recipe wants an eighth of the block. An eighth is found by halving three times: $64 \rightarrow 32 \rightarrow 16 \rightarrow 8$. The decimal route needs 0.125×64 , a three-place decimal multiplication, which is more work by hand for the same answer. The recipe calls for 8 grams.

10. Two percents in a row: 60 dollars, 25% off, then 8% tax. The discount is $\frac{25}{100} \times 60 = 15$ dollars, so the equation for the sale price is $s = 60 - 15$, giving $s = \span style="border: 1px solid black; padding: 0 2px;">45 dollars. Tax is charged on what the shop actually sells the backpack for, which is the sale price, so the tax is $\frac{8}{100} \times 45 = 3.6$ dollars. Final price: $45 + 3.6 = 48.6$. To the nearest cent the final price is 48.60 dollars.$

Taking the tax on the original price would give $60 + 4.8 = 64.8$ dollars, which overcharges by more than the discount was worth — the tax rate applies to the amount being paid, not to a price nobody pays.